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Remembrance of Places Passed: Spatial Memory in Rats

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Rats were tested on an eight-arm maze in a paradigm of sampling with replacement from a known set of items until the entire set was sampled. The first three experiments demonstrated that the animals performed efficiently, choosing an average of more than seven different arms within the first eight choices, and did not utilize intramaze cues or consistent chains of responses in solving the task. The second three experiments examined some characteristics of the rats' memory storage. There was a small but reliable recency effect with the likelihood of a repetition error increasing with the number of choices since the initial instance. This performance decrement was due to interference from choices rather than just to the passage of time. No evidence was found for a primary effect. The data also suggest that there was no tendency to generalize among spatially adjacent arms. The results are discussed in terms of the memory processes involved in this task and human serial learning.

When distinctive exteroceptive discriminative stimuli are consistently associated with a particular spatial location, rats preferentially use these stimuli for discrimination learning, a phenomenon that is usually referred to as "place learning." If place learning can be used to solve a discrimination problem, rats learn very rapidly. If place learning cannot be used to solve a discrimination problem, rats learn slowly and almost invariably adopt a "position habit" or "spatial hypothesis" before finding the correct solution. (Relevant literature reviews may be found in Gleitman, 1955; Kimble, 1961, p. 223; Olton & Samuelson,

1974; Sutherland & Mackintosh, 1971; Tolman, Ritchie, & Kalish, 1946; Woodworth, 1938, p. 124). Also, spontaneous alternation in a T maze is strongly influenced by spatial cues (Douglas, 1966), and during avoidance behavior rats preferentially associate shock with spatial location rather than with visual or auditory cues (Olton, 1973).

In spite of the ubiquitous nature of place learning, most experiments have treated place learning as a factor to be controlled and have chosen to assess rats' cognitive abilities by making place learning impossible. In contrast, the present experiment was designed to permit place learning and to utilize this learning to assess the capacity of rats to discriminate, remember, and process information derived from place learning in searching for food. To accomplish this goal, we developed a new experimental testing paradigm, one which requires sampling with replacement from a known set of items until the entire set is sampled. In order to establish this procedure, an appa-

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ratus was constructed which required the rat to return to the same central choice point after each choice so that the entire set of choice alternatives (including the ones already chosen) was presented on each trial. The reinforcement contingencies were arranged so that the first choice of each alternative was reinforced, while subsequent choices were not, making the optimal behavior choosing each alternative once until all alternatives were chosen. This procedure is in distinct contrast to the typical maze learning task in which choice alternatives are presented sequentially and the animal is faced with only one subgroup (usually a pair) of the entire set of alternatives at any given time.

The first three experiments were designed to investigate the general characteristics of performance in this paradigm and to assess the relative importance of extramaze cues, intramaze cues, and response chains in order to determine the type of strategy used by the rats. The subsequent experiments were then conducted to describe the characteristics of the memory processes that underlie performance.

EXPERIMENT 1

Method

Subjects. The subjects were six experimentally naive male albino rats that weighed 300–325 g at the start of testing.

Apparatus. The apparatus was a radial eight-arm maze as diagrammed in Figure 1. The center platform was 34 cm wide. All arms were the same length (86 cm) but various widths (1.3, 2.5, 3.8, 5.0, 6.3, 7.5, 9.3, and 10.0 cm). The entire apparatus was made of wood and was approximately 50 cm above the floor. The testing room was well illuminated by ceiling lights unless otherwise noted. There were numerous visual cues scattered around the room, including a sink, table, columns, and so on.

Procedure

Exploration. During the first 9 days all rats had ad-lib food and water in the home cage. On each day they were given approximately 15 min. to explore the apparatus; no food was present in the apparatus during this time.

Free choice. For the rest of the experiment, all rats were placed on a food deprivation schedule to bring body weights down to 85% of ad lib; daily feeding took place within $\frac{1}{2}$ hr. after testing. During the next 10 days (Days 10–19), reinforce-

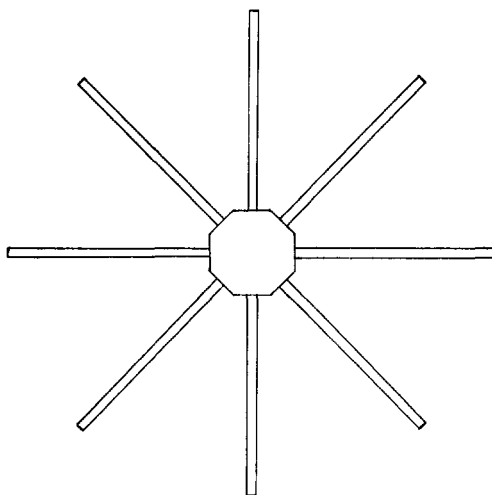


FIGURE 1. Diagram of a top view of the apparatus.

ment was provided by pieces of Purina Lab Chow that weighed approximately .1 g each. At the start of each test session, one piece of food was placed at the end of each arm; reinforcement was not replaced during the test so that a maximum of eight pieces of food could be obtained. Each rat was placed on the center platform of the apparatus and allowed to make 16 choices or given a total of 10 min. if 16 choices were not made. A choice was defined as the rat proceeding to the end of an arm.

Rebait. During the next 5 days (Days 20–24), reinforcement was replaced on one of the eight arms after the rat had chosen that arm and returned to the center platform. The actual arm rebaited varied, but in all cases it was an arm chosen within the first six choices. If the presence of food at the end of an arm influenced the rat's choice of that arm, the rat ought to return to the rebaited arm after fewer choices than in the free choice procedure.

Added odor. During the next 3 days (Days 25–27), the apparatus was liberally sprinkled until wet with either Old Spice or Mennen aftershave lotion. The lotion was applied to both the center platform and to all of the arms just prior to testing each animal. If the odor of food at the end of an arm or some kind of "odor trail" within the apparatus influenced the rat's choice behavior, then choice accuracy here should be lower than that in the free choice procedure.

Results

Exploration. During the first day of testing, all rats moved freely about the center platform of the apparatus but rarely went onto the arms. Over the next few days they began to venture out onto the arms until they would readily run from the center plat-

form to the end of an arm and back again. Typically, the rats would pause at the edge of the platform before entering an arm and orient toward the end of the arm while rapidly moving their vibrissae. This orienting behavior was followed either by withdrawal to the center of the apparatus and repetition of the orienting behavior at another arm or by a choice of the arm. On other occasions, the rats would run rapidly across the center platform and without pause run down an arm; this type of behavior was usually observed only on the first few choices. In almost all cases, once a rat had placed all four legs onto an arm, it proceeded to the end of the arm before turning around. There was a marked preference for the wider arms, and rats rarely chose the three most narrow arms.

Free choice. The general activity level of the rats increased markedly from the preceding exploration conditions. At first, some food pellets were left on an arm even though the rat had placed its nose very close to the pellet and presumably noticed its presence. Within a few days, all food pellets were taken from the arms and consumed, usually in the center platform. As the rats became more familiar with the procedure, they would often eat either at the end of an arm or while running along the arms and making subsequent choices. As before, rats would often pause at the edge of the center platform and orient toward the end of an arm before choosing it, and once the rat had all four legs on the arm, it almost always proceeded to the end before turning around.

All rats rapidly came to choose an average of more than seven different arms within the first eight choices. The mean numbers of different (i.e., correct) arms chosen within the first eight choices for the first 5 and second 5 days of this procedure were 5.7 (range: 4.2–6.6) and 7.6 (range: 7.2–8.0), respectively.

Rebait. Replacing food on one of the arms after the original piece of food was obtained had no obvious effect on the accuracy of choice behavior. Occasionally, the animals seemed to spend more time than usual sniffing and orienting after the food pellet was

replaced, but this behavior was not reliable. No rat ever returned to the rebaited arm before making at least eight choices, and on 5 of the 30 tests (six rats, five tests each) the rebaited arm was never chosen within the 16-choice limit. The mean number of choices before returning to the rebaited arm was 11.6 (range: 9.6–13.0). Since the rats tended to choose all eight arms again during Choices 9–16, this performance is very similar to that expected by chance, indicating that the presence of food at the end of an arm had no observable effect on the accuracy of choice behavior.

Added odor. The addition of the after-shave lotion had a profound effect on the animals' general behavior. On the first day, all animals were greatly disturbed by the odor; some were very hesitant to run on the arms and spent most of the time on the center platform, while others would often go out to the end of an arm but refuse to pick up the pellet. By the second day, this behavior pattern was greatly diminished, and all animals ran readily down the arms and consumed all the pellets. In contrast to the disruption of general behavior, there was little change in the accuracy of choice behavior even on the first day. The mean number of correct choices within the first eight choices was 7.4 (range: 7.3–8.0) over all three days, indicating that the odor from food or some kind of odor trail had no observable effect on choice accuracy.

All procedures. To simplify the analysis of choice behavior during Choices 9–16, only those days on which the animals both (a) obtained all eight pieces of food within the first eight choices and (b) continued to make a total of 16 choices were analyzed. For the six rats, there was a mean of 9.2 (range: 3–11) days on which both Conditions a and b were met. On these days, the mean number of different arms chosen during the last eight choices was 5.9 (range: 4.3–7.2). Thus, even after all the food had been obtained from the apparatus, the rats still tended to choose each of the arms, although not to the extent as during Choices 1–8. A similar pattern of choice behavior was observed on days when Conditions a and b were not met.

The order of choices was analyzed to determine if the rats used specific sequences to simplify the problem. During all of the procedures, the rats almost never chose the same arm twice in succession and this sequence was found in less than .01% of the choices. In general, the rats tended not to choose adjacent arms, although occasionally there were sequences in which two, three, or four adjacent arms were chosen in order. The rats often responded in a clockwise or counterclockwise fashion and seemed to prefer the arm 90° from the arm just chosen. None of these patterns or any other pattern obvious to the experimenters was regularly exhibited by the rats, and, most importantly, choice accuracy was routinely high irrespective of the particular order of choices. Table A1 presents the raw data for Rat 3 during Days 10–27. Although various patterns of choices appear, there are no patterns that are a prerequisite for accurate choice behavior.

Discussion

In the typical sequential maze experiment, decision making at any given choice point can be independent of that at any other choice point. The rat must decide between only one pair of alternatives (i.e., go left, go right) at a time, and this choice can be made without reference to any preceding or subsequent choices. In the present experiment, however, the rat always returned to the center platform after each arm and was thus presented simultaneously with all the alternatives on each choice. Thus the rat cannot treat each successive choice independently. The paradigm may be summarized as one which requires sampling with replacement from a known set of items until the entire set is sampled.

Rats performed very accurately and within 5 days of free choice testing, all animals were consistently choosing an average of more than seven different arms within the first eight choices. The high level of accuracy even on the first 5 days of free choice testing suggests that the rats were predisposed not to repeat their choices in this situation, presumably reflecting the tendency toward spontaneous alternation commonly observed in two-choice paradigms in a simple T maze

(Douglas, 1966). Accurate determination of "chance" performance levels given this alternation tendency is difficult. If each choice is completely independent of all others, the experiment becomes an example of the classic occupancy problem (Feller, 1950); the probabilities of choosing n different arms within the first eight choices are as follows: $n = 8$, $p = .0024$; $n = 7$, $p = .0673$; $n = 6$, $p = .3194$; $n = 5$, $p = .4200$; $n = 4$, $p = .1680$; $n = 3$, $p = .0168$; $n = 2$, $p = .0004$; $n = 1$, $p = 6 \times 10^{-8}$. The assumption of choice independence for rats is probably not valid because even on the first few days of exploration the rats almost never chose the same arm twice in succession and usually chose at least five different arms within eight choices. In another experiment (Olton, Note 1) infant rats were rewarded for a response to any arm, as in the typical reinforced spontaneous alternation paradigm. Under these conditions, the animals chose an average of six different arms within the first eight choices, indicating a strong disposition not to repeat choices to arms even when food was obtained on repeated arms as well as on new arms. The presence of a spontaneous alternation tendency does not compromise the basic argument about the discrimination requirements of the paradigm made above because in order to exhibit spontaneous alternation, the rat must obviously locate each arm and determine which arms have been chosen and which have not. The only difficulty imposed by the presence of spontaneous alternation tendency is accurate assessment of the relative contribution of learning and of innate behavioral tendencies to successful performance, and the present experiment was not designed to resolve this issue.

Rebaiting the arms with food and adding odor to the arms did not markedly influence the choice accuracy of the animals, demonstrating that cues such as the odor or sight of food at the end of an arm and odor trails from the animals themselves were unimportant in performing the task. A number of studies have demonstrated that odor can function as a cue for rats in spontaneous alternation (Douglas, 1966), maze learning (Means, Hardy, Gabriel, & Uphold, 1971;

Wasserman & Jensen, 1969) and other types of tasks (Slotnick, 1974). In all of the above experiments, however, the salience of the odor was markedly increased either by combining odors from many rats all tested under the same condition or by having all odors except the stimulus odor removed or suppressed. The data from the present experiment suggest that under more natural conditions where the odor trail from a single rat is deposited among the odor trails from many other rats exhibiting different behaviors, odor trails have a minimal effect on discrimination behavior.

The number of different arms chosen during Choices 9–16 was slightly less than that during Choices 1–8 but was still routinely high, suggesting that animals tended to choose each arm even after all the food had been obtained. The results of Experiment 6 to be reported below suggest that the slight decrease in the number of arms chosen during the second eight choices probably reflects the performance decrement that would be expected in the absence of differential reinforcement. The results offer further evidence for the unimportance of intramaze odor cues because after the first eight choices the rat would have left an odor trail on all arms, while there would be no food on the end of any arm; nonetheless, the number of different arms chosen was still high.

Rats have been shown to "chain" specific responses to form very long specific behavioral sequences (Hulse, Deese, & Egeth, 1975, p. 56). In the present experiment, however, no evidence of chaining was observed. Although the rats did occasionally repeat specific sequences of choices, these were not regularly observed and choice accuracy was routinely high irrespective of the particular choice pattern emitted. The data indicate that performance here is very different from that observed in behavior chains and that chaining was not utilized as a strategy to solve the task.

In summary, rats perform well in the present situation, choosing almost perfectly during their first eight choices. Intramaze odor cues and consistent sequences of choices are both unnecessary for successful performance.

EXPERIMENT 2

The present experiment was designed to provide additional evidence concerning the relative importance of intramaze cues, extramaze cues, and choice sequences. To this end, all arms were made equally wide so that arm width could not serve as a cue; animals were confined to the center platform by means of guillotine doors in order to interfere with any sequence of choices that might have been in process at this time; arms were interchanged so that all extramaze place cues were dissociated from all intramaze cues.

Method

Subjects and apparatus. The subjects were the same as in Experiment 1. The apparatus of Experiment 1 was modified in the following manner. All arms were 7 cm wide. A hole 2 cm wide and 1 cm deep was drilled at the end of each arm to serve as a food cup. The arms were made so that they could be detached from the center platform and interchanged with each other. A wooden compartment 10 cm high was constructed around the edge of the center platform. Wooden guillotine doors were placed in the walls of this compartment at the entrance to each arm so that the animal could be confined to the center platform. The top of the compartment was of Plexiglas.

Procedure

Free choice. For 10 days (Days 1–10) the rats were tested as in the free choice procedure of Experiment 1. They were allowed either 16 choices or 10 min. in the apparatus, whichever came first.

Confinement. For five days (Days 11–15) the rats were confined to the center compartment by lowering the guillotine doors when they entered the center compartment after their third choice. The doors remained lowered for approximately 1 min., after which they were raised and the rat was allowed to choose freely until a total of 16 choices had been made or 10 min. elapsed.

Interchange arms. For 15 days (Days 16–30) the rats continued to be confined to the center compartment after their third choice. During this confinement, two adjacent arms were interchanged. The actual arms moved varied each day, but in all cases one arm had been chosen during the first three choices and the other arm had not, placing spatial location cues and arm cues in opposition to each other. In the following discussion, "chosen arm" refers to the arm of the interchanged pair which had been chosen prior to the arms being interchanged, whereas "unchosen arm" designates the arm of the interchanged pair which had not been chosen prior to the arms being interchanged. Likewise, "chosen spatial location" refers to the

position in the testing room of the arm of the interchanged pair which had been chosen prior to the arms being interchanged, and "unchosen spatial location" indicates the position in the testing room of the arm of the interchanged pair which had not been chosen prior to the arms being interchanged. To the extent that arm cues (such as odor trail) direct choice behavior, the rats should go to the unchosen arm while repeating their choice of spatial location. To the extent that spatial location cues direct choice behavior, the rats should go to the unchosen spatial location while repeating their choice of the arm. Three rats were reinforced for responding to the unchosen arm following confinement, while the other three rats were reinforced for responding to the unchosen spatial location following confinement. Reinforcement on the remaining six arms was not altered.

Results

Free choice. Making all arms the same width had no obvious effect on the accuracy of choice behavior. The mean number of different arms chosen during the first eight choices was 7.7 (range: 7.4–7.9).

Confinement. While confined to the center compartment the animals were usually active, exploring and moving about the compartment for the duration of the confinement period. When the guillotine doors were opened, the rat usually approached and oriented toward the end of the arm it was facing. The accuracy of choice behavior was not at all affected by the confinement; the mean number of different arms chosen in the first eight choices was 7.7 (range: 7.0–8.0).

Interchange arms. In all cases, the animals showed little, if any, evidence of reaction to the interchanged arms and continued to choose almost entirely on the basis of spatial location. The rats reinforced for responding to the unchosen arm performed no differently from the rats reinforced for responding to the unchosen spatial location and the data from both groups were combined for analysis. In all but 2 of the 90 tests (6 rats, 15 tests each), rats responded to the unchosen spatial location prior to responding to the unchosen arm. The mean choice number of the unchosen spatial location was 6.1 (range: 4.1–7.3) and was thus always among the first eight choices when food was usually obtained, while the mean choice number of the unchosen arm was 9.9 (range: 8.5–11.7) and thus always among the second eight

choices when food was usually not obtained. The accuracy of the first eight choices for spatial location did not decrease as a result of interchanging arms, and the mean number of different spatial locations chosen within the first eight choices was 7.7 (range: 7.6–7.8).

All procedures. The order of choices was again analyzed. As in the previous experiment, various patterns often appeared on a particular day, but none of these patterns was necessary for accurate choice behavior. The rats also tended to choose each of the eight different arms on Choices 9–16.

Discussion

The results of Experiment 2 extend and corroborate those of Experiment 1 describing the types of cues directing the rats' choice behavior. First, the unimportance of intramaze cues was supported by continued high performance when all arms were changed to equal width. Second, the unimportance of specific continuous responses sequences was supported by continued high performance when animals were confined to the center compartment between choices. Third, the unimportance of intramaze odor cues was strongly corroborated by interchanging the arms. Those animals reinforced for responding to the unchosen spatial location were unaffected by the presence of an odor trail and those animals reinforced for responding to an odor trail (i.e., the unchosen arm) showed no evidence of learning to follow the odor trail.

EXPERIMENT 3

Experiment 3 was designed to replicate and extend the results of the previous experiments in three ways. First, the apparatus was altered so that instead of interchanging just two arms and leaving any possible intramaze cues on the center platform unaltered, the entire arm assembly could be rotated while the animal was in the center compartment, and then the entire center compartment rotated while the animal was on an arm, thus rotating all intramaze cues instead of just those on two arms. Second, no exploration period took place prior to training to obtain a better estimate of the rate of

learning. Third, new animals were used to determine the reliability of the previous findings.

Method

Subjects. The subjects were six experimentally naive male albino rats that weighed 300–325 g at the start of testing. All rats were food deprived to 85% of ad-lib body weight and were maintained at this level throughout the experiment by feeding the required amount of Purina Lab Chow within $\frac{1}{2}$ hr. after testing.

Apparatus. The apparatus was the same as in Experiment 2 except that the arms and center compartment were suspended independently of each other so that each could be rotated without disturbing the other. Reinforcement consisted of one 190-mg Noyes food pellet at the end of each arm.

Procedure

Free choice. For the first 15 days, each rat was given one daily test session until 16 choices were made or a total of 10 min. had elapsed. On the first few days, rats were shaped as necessary to go onto the arms by placing food reinforcement close to the center platform. On Days 6–15, all food was placed in the food cups at the end of the arms.

Confinement. For the next five days (Days 16–20), each rat was confined in the center compartment for approximately 1 min. after the third choice by lowering the guillotine doors after the rat entered the center compartment.

Maze rotation. For the next six days (Days 21–26), each rat was confined to the center compartment after the third choice as described above. On Days 1, 3, and 5, while the rat was confined to the center compartment which did not move at this time, the arm assembly was rotated 45° and food was replaced on the three arms already chosen on that day so that all arms contained food. The guillotine doors were then raised and, while the rat was on the arm making its fourth choice, the center compartment was rotated 45° in the same direction as the arm assembly. Thus, all intramaze cues were rotated 45° with respect to extramaze cues. The rat was allowed to choose freely until either the five remaining arms or the five remaining spatial locations were chosen. On Days 2, 4, and 6, each rat was confined after the third choice and the maze arms were rotated, but the arms were returned to their original location before the guillotine doors were raised in order to control for any general influence of the movement of the apparatus on choice behavior.

Results

Free choice. For the first few days, all rats were reluctant to enter the arms and

spent considerable time exploring the center compartment and orienting toward the end of the arms. Within 8 days, all animals readily ran on the arms, retrieving and consuming all the food pellets. Their general behavior was similar to that reported previously; the rats often hesitated at a guillotine door and oriented toward the end of the arm. Once all four legs were placed on the arm, however, the rat almost always ran to the end of the arm and placed its nose in the food cut. No data are reported for the first 5 days because the rats often did not leave the center compartment to run down the arms. As soon as the rats began to run and choose all the arms, they very rapidly demonstrated the discrimination; the mean numbers of different arms chosen within the first eight choices for Days 6–10 and Days 11–15 were 7.2 (range: 6.8–7.4) and 7.3 (range: 6.8–7.8), respectively.

Confinement. Confinement had no obvious effect on the animals' behavior. As before, the animals were active during the period of confinement, exploring the center compartment; when the guillotine doors were open, they usually oriented down the arm they were facing. The mean number of different arms chosen within the first eight choices during this period was 7.6 (range: 7.5–7.8).

Maze rotation. To the extent that intramaze cues control choice behavior, the rats ought to go to the five remaining unchosen arms after maze rotation and repeat choices to already chosen spatial locations. To the extent that extramaze cues control choice behavior, the rats ought to go to the five remaining unchosen spatial locations after the maze rotation and repeat choices to already chosen arms. Consequently, the relevant data are the number of times an already chosen arm was repeated after the maze rotation and the number of times an already chosen spatial location was repeated after the maze rotation. Since three choices were made prior to the maze rotation, the maximum number of repetitions was three. The mean number of repetitions of already chosen arms was 2.3 (range 1.3–2.7), while the mean number of repetitions of already chosen spatial locations was .3 (range: .0–1.0). Additionally, the mean number of different spatial loca-

tions chosen within the first eight choices was 7.5 (range: 7.2–8.0) on days when the maze was rotated, the same values as on control days.

Discussion

Results of Experiment 3 are consistent with those of the previous two experiments. After the animals learned to run out onto the arms, their choice accuracy was very high. Manipulation of intramaze cues appeared to have no effect on choice behavior; all animals continued to choose accurately on the basis of spatial location and readily repeated already chosen arms in order to respond to unchosen spatial locations.

EXPERIMENT 4

The results of Experiments 1, 2, and 3 consistently demonstrate that the rats did not choose adjacent arms, did not rely on any type of intramaze marking, did not chain responses in particular sequences, and did not utilize any other obvious strategy to simplify the task. Since the testing procedure involved sampling with replacement, the rats must have been able to identify each of the eight arms and remember over a period of several minutes which arms had been chosen and which had not. In short, the rats seemed to treat each of the eight spatial locations separately.

The purpose of the present experiment was to investigate the characteristics of the memory for this list of items. In particular, the following questions were addressed: (a) Is there a performance decrement as the number of choices the animals makes increases? (b) Are correct choices at the beginning of the test remembered better than subsequent choices? (c) Are correct choices at the end of the test remembered better than previous choices? (d) Is there spatial generalization from one arm to the other so that errors tend to be directed towards correct choices? (e) Is information from correct choices and incorrect choices stored in the same or different memory stores?

Method

Subjects and apparatus. The subjects were 12 experimentally naive male albino rats that weighed

300–325 g at the start of testing. The apparatus was the same as in Experiment 3.

Procedure

Rats were placed on a food deprivation schedule to bring body weights down to 85% of ad lib; daily feeding took place approximately $\frac{1}{2}$ hr. after testing. Reinforcement was a 45-mg Noyes pellet. At the start of each test session, one pellet was placed at the end of each arm; reinforcement was not replaced during the test so that a maximum of eight pieces of food could be obtained. Each rat was placed on the center platform of the apparatus and allowed to make choices until all eight reinforcements were obtained or until 10 min. had passed, whichever came first. All rats were tested for 30 days, with one test on each day.

Data Analysis

The data analysis for this experiment will be presented in considerable detail because there are a number of different factors that must be considered and there are no readily available precedents to use as a model for analysis. For each question, two types of analyses will be presented. The first is designed to describe the actual performance of the animal and will provide information about the observed behaviors. The second is designed to adjust for various influences which must be considered in order to adequately answer the five questions posed in the introduction. In all cases, the procedure chosen for the second analysis is the one that seems most appropriate to the question being asked. But for any set of raw data and any particular experimental question, a number of different analyses are possible, each based on a different set of assumptions and with its own set of advantages and disadvantages (e.g., see Murdock, 1957). Thus, the first analysis not only provides information about the way in which the animals actually performed but also allows the reader to carry out other transformations of the data if desired.

All but one of the analyses are based on errors (i.e., returning to an already chosen arm). As in the previous experiments, the animals performed well and made relatively few errors. In order to obtain enough observations for an accurate estimate of behavior, the data for each animal were averaged over several days. Since choice accuracy gradually improved during the first 10 days and then remained stable during the second 10 days, the data were analyzed in 10-day blocks. Thus, the results from the first 10 days probably reflect some learning of the task, while the results from the second 10 days probably reflect asymptotic performance of the already learned task. During Days 11–20, one animal performed perfectly so that there were no errors to analyze. Four other animals had either one or two errors, whereas the remaining seven had four or more errors. For Days 11–20, the data were analyzed separately for $n=11$ and for $n=7$.

Analyses of variance were for a treatments by subjects design.

Results

A mean of 4.9 days (range: 3–8) was required before a total of eight choices were made within the 10-min. test limit. On subsequent days, animals chose readily, always completing at least eight choices and usually finishing the day's test within 2 min. For purposes of data analysis, Day 1 was defined as the first day on which at least eight choices were made.

Choice accuracy: General. The task was learned rapidly and even during Days 1–5 the mean number of different arms chosen within the first eight choices was 6.9 (range: 6.0–7.3). Performance improved to a mean of 7.5 (range 6.9–7.8) correct responses during Days 6–10 and then appeared to reach an asymptote at about 7.6 (range: 7.3–8.0) correct responses within the first eight choices on Days 11–15 and Days 16–20. All but one animal performed better during Days 16–20 than during Days 1–5, indicating that although initial performance was accurate, a significant amount of learning did take place.

Performance decrement: Observed probability of a correct response. Performance accuracy was greatest at the beginning of each day's test and declined slightly but consistently during the test. Table 1 presents the means for the observed probability of a correct response, $p(\text{cor})_{\text{obs}}$, on Choices 2–8 during Days 1–10 and Days 11–20. Data for the first choice are not presented because the first response had to be correct. Data for choices following Choice 8 are not presented because the animals often performed perfectly and testing was terminated at the end of 8 choices. The observed probability of a correct response was determined for each animal for each choice by the formula

$$p(\text{cor})_{\text{obs}} = \frac{\text{number of correct responses}}{\text{total number of responses}} \times 100.$$

The $p(\text{cor})_{\text{obs}}$ score can range from 100, which indicates a correct response on all choices, to 0, which indicates an incorrect response on all choices. As would be ex-

TABLE 1
PROBABILITY OF A CORRECT RESPONSE

Probability	Choice						
	2	3	4	5	6	7	8
Observed							
Days 1–10	100	95	95	85	87	81	78
Days 11–20	100	99	98	96	97	90	80
Transformed							
Days 1–10	100	84	86	71	77	70	73
Days 11–20	100	96	95	91	95	88	77

pected from the data reported in the previous paragraph, $p(\text{cor})_{\text{obs}}$ was high on all choices. Scores were greatest on the second choice and steadily declined until the eighth choice (Days 1–10, $F(6, 66) = 5.1$, $p < .01$; Days 11–20, $F(6, 66) = 9.6$, $p < .01$). Scores for all choices were greater during Days 11–20 than during Days 1–10, reflecting the improvement in accuracy reported in the previous paragraph as measured by the number of correct responses within the first 8 choices.

Performance decrement: Transformed data. Although the observed probabilities in Table 1 are clear, there is a bias present which tends to exaggerate the magnitude of the decline in performance during the test. This bias arises because the chance probability of a correct response decreases as the number of arms already chosen increases. For example, consider a sequence of choices in which all choices are correct; on the second choice, seven of the eight arms still contain food, on the third choice, six of the arms contain food, and so on. Thus any given $p(\text{cor})_{\text{obs}}$ score may reflect relatively better performance after many choices than it does after few choices, and the observed probability data tend to exaggerate the decline in performance that occurs during each test. To correct for this bias, the changing expected chance probability of a correct response must be considered. To this end, $p(\text{cor})_{\text{exp}}$ was defined as the expected chance probability of a correct response on any given choice according to the formula

$$p(\text{cor})_{\text{exp}} = \frac{\text{number of arms not chosen}}{8} \times 100.$$

The observed probability data from Table 2 were then transformed for each animal for each choice, according to the formula below, to provide a measure of the extent to which the animals exceeded chance performance and attained maximum performance on each choice:

$$\frac{p(\text{cor})_{\text{obs}} - p(\text{cor})_{\text{exp}}}{100 - p(\text{cor})_{\text{exp}}} \times 100.$$

This formula corrects for the expected chance performance on each choice and expresses performance as a proportion of the maximum performance obtainable, allowing performance on any given choice to be directly compared with performance on any other choice. The transformed scores can range from 100, indicating the maximum number of correct responses, through 0, indicating random performance with $p(\text{cor})_{\text{obs}} = p(\text{cor})_{\text{exp}}$, to -100, indicating the maximum number of incorrect responses. [When $p(\text{cor})_{\text{obs}}$ was less than $p(\text{cor})_{\text{exp}}$, the denominator was changed to $p(\text{cor})_{\text{exp}} - 0$.] For example, consider an animal with $p(\text{cor})_{\text{obs}} = 80$. If this performance was on the third choice and six arms had not been chosen, the transformed score would be $(80 - 75)/(100 - 75) \times 100 = 20$. If this performance was on the seventh choice and two arms had not been chosen, the transformed score would be $(80 - 25)/(100 - 25) \times 100 = 73$. As the number of correct arms available to the animal decreases, the transformed score for any given $p(\text{cor})_{\text{obs}}$ increases. Also, as the accuracy of performance increases, the transformed score approaches the asymptote of 100.

Table 1 also presents the transformed scores. An important result is that all scores were substantially greater than 0, indicating greater than chance performance on even the last choice. The same trends that were apparent in the observed probabilities are also apparent here. First, performance gradually decreased as the number of choices increased (Days 1-10, $F(6, 66) = 6.5$, $p < .01$; Days 11-20, $F(6, 66) = 7.1$, $p < .01$). Second, performance during Days 11-20 was better than performance during Days 1-10.

Sequential position of errors: General analysis. The data in Table 1 indicate that

the likelihood of making an error increased with the number of choices made by the animal, but they do not provide information about the sequential characteristics of these errors. When errors did occur, they were almost always to arms that had been chosen at the beginning of the day's test and rarely to arms chosen near the end of the day's test. Four different analyses of these data were performed to provide a quantitative measure of this behavior: (a) the observed relative probability of repeating each of the first seven correct choices, $p(\text{rep})_{\text{obs}}$; (b) correction of $p(\text{rep})_{\text{obs}}$ for the differing number of opportunities to repeat each choice; (c) correction of $p(\text{rep})_{\text{obs}}$ for the possibility of an arm preference; (d) correction of $p(\text{rep})_{\text{obs}}$ for both differing number of opportunities to repeat each choice and the possibility of an arm preference.

In all of these analyses, the results were the same. The probability of an error was greater for arms chosen early during each day's test than for arms chosen later during each day's test (for Days 1-10 and Days 11-20, $ns = 7$ and 11, respectively). Analyses of variance yielded F s ranging from 2.7 to 15.9 ($p < .05$ for 2 tests, $p < .01$ for 10 tests). To conserve space, each analysis is described, but only the data from Days 1-10 are presented in Table 2. Table A2 presents the raw choice data upon which the analyses are based.

Sequential position of errors: Observed relative probability of repetition. The observed relative probability of repeating each of the first seven correct choices was determined. The eighth choice was not considered because testing was terminated after this choice and there was no opportunity to repeat the eighth choice. For each animal for each day, all correct choices were ordered in sequence from one to seven. Then for each animal, the observed relative probability of repeating each correct choice was calculated according to the formula

$$p(\text{rep})_{\text{obs}} = \frac{\text{number of repetitions of choice}}{\text{total number of repetitions}} \times 100.$$

The $p(\text{rep})_{\text{obs}}$ score indicates the relative probability of an error to each of the previous

TABLE 2

RELATIVE PROBABILITY OF REPEATING A PREVIOUSLY CORRECT CHOICE: DAYS 1-10

Analysis	Correct choice						
	1	2	3	4	5	6	7
Observed probability	36	23	15	12	8	6	1
Correction for opportunities	43	30	23	23	19	21	5
Correction for arm preference	31	23	17	12	9	6	.8
Correction for opportunities and arm preference	40	31	27	26	22	25	8

correct choices. Scores can range from 100, indicating that all errors were made by repeating one particular choice, to 0, indicating that no errors were made by repeating that choice. The sum of scores for all choices equals 100. Consider an animal that made a total of 8 errors in 10 days. If 5 of these errors were to the first correct choice, 2 to the second correct choice, and 1 to the third correct choice, then the $p(\text{rep})_{\text{obs}}$ score for correct Choices 1, 2, and 3 would be 63, 25, and 13, respectively. The first line of Table 2 presents the mean of $p(\text{rep})_{\text{obs}}$ scores for Choices 1-7 during Days 1-10. As can be seen from Table 2, when errors occurred, they were most likely to be to the first correct choice, and there was a steady decline in the probability of repetitions for each subsequent choice.

Sequential position of errors: Adjustment for opportunities. There is a bias in the data, as presented in the first line of Table 2, because the number of opportunities available for repeating each choice differs. The animals made an average of 9.3 choices to obtain all eight pieces of food during Days 1-10 and, thus, had an average of 8.3 opportunities to repeat the first correct choice and a maximum of 7.3 opportunities to repeat the second choice, 6.3 for the third, and so on. An adjusted score, correcting for the number of opportunities to repeat each choice, was calculated for each animal for each choice according to the formula

$$\frac{p(\text{rep})_{\text{obs}}}{\text{number of choices remaining}} \times 100.$$

Unlike $p(\text{rep})_{\text{obs}}$, these adjusted scores do not sum to 100 and range from 0 to greater

than 100. The adjusted scores can be 0, indicating that no errors were made to the choice in question, or they can be some positive number, indicating that some errors were made to that choice: The greater the number, the greater the number of errors made to that choice. These adjusted scores also correct for the number of opportunities to repeat each choice and indicate the likelihood of repeating a choice any time an error is made. Consider an animal with $p(\text{rep})_{\text{obs}} = 40$ for the first 10 days. If this score was obtained for the first choice of each day and there was a total of 80 choices following the first choice (indicating that the animal obtained all eight pieces of food in an average of 9.0 trials), then the adjusted score would be $40/80 \times 100 = 50$. If this score was obtained for the third choice of each day and there were 50 choices remaining (indicating perfect performance on Choices 2 and 3), then the adjusted score would be $40/50 \times 100 = 80$. For any given $p(\text{rep})_{\text{obs}}$ score, the adjusted score increases as the number of choices remaining decreases. The second line of Table 2 presents these adjusted scores.

Sequential position of errors: Adjustment for arm preference. If a strong preference for one or more arms developed, the animal might tend to choose this arm early during each day's test. If the arm preference were constant throughout the day's test, then the animal might continue to have a strong tendency to respond to this arm. Such a combination of events would lead to the animal repeating early choices, not because of the position of the choice within the sequence but rather because of a preference for the arm. An attempt to evaluate and correct for any possible arm preference was undertaken by determining for each arm the average sequential position of the first correct response to that arm on each day. Each first choice of an arm was given a score of 1, each second choice, 2, and so on. All these scores were summed and divided by 10 for each 10-day block. The resulting sequential position score provides for each arm the average sequential position of the first correct response to that arm during the 10 days. For each animal, the scores must total 36 (i.e., $1 + 2 + \dots + 8$). The expected chance performance

score for any arm is 4.5 (i.e., $[1 + 2 + \dots + 8]/8$). A score of less than 4.5 indicates a tendency to choose an arm preferentially early in the day's test; a score of more than 4.5 indicates a tendency to choose an arm preferentially late in the day's test. All animals exhibited a preference for some arms over others. The mean (and standard error of the mean) sequential position score of the most preferred arm was $2.8 (\pm .06)$ and of the least preferred arm was $6.0 (\pm .04)$. In order to determine how stable any arm preference might be, a correlation coefficient was calculated for each animal between the preference scores on each of the eight arms for Days 1-10 and the preference scores on each of the eight arms on Days 11-20. The correlations ranged from $-.63$ to $.89$, with an average of $.46$.

The $p(\text{rep})_{\text{obs}}$ scores of the first line of Table 2 were adjusted for arm preference by using sequential position as a weighing factor, multiplying each arm by its sequential position score, according to the formula

$$\frac{\text{sequential position score for choice}}{\text{sum of sequential position scores for all choices}} \times 100.$$

Like $p(\text{rep})_{\text{obs}}$, the adjusted scores can range from 100, indicating that all errors were made by repeating that particular choice, to 0, indicating that no errors were made by repeating that particular choice, and the sum of scores for all choices is 100. Unlike $p(\text{rep})_{\text{obs}}$, the adjusted scores correct for the relative preference for each arm as measured by the sequential position score. Consider an animal that repeats choices to Arms 1, 4, and 5 which have sequential position scores of 1.5, 4.0, and 4.5, respectively, and makes two repetitions of the first choice, both of Arm 1, and two repetitions of the second choice, one each to Arms 4 and 5. The adjusted score for the first choice is $[(1.5 + 1.5)/11.5] \times 100 = 27$, the adjusted score for the second choice is $[(4.0 + 4.5)/11.5] \times 100 = 86$, and the adjusted scores for all other choices is 0. [The unadjusted $p(\text{rep})_{\text{obs}}$ scores reported as in Table 1 would be: Choice 1, $2/4 \times 100 = 50$; Choice 2, $2/4 \times 100 = 50$; all other choices = 0.] Thus for

any given $p(\text{rep})_{\text{obs}}$ score, the greater the preference for an arm as measured by the sequential position score, the lower will be the adjusted score. Line 3 of Table 2 presents these adjusted scores.

Sequential position of errors: Adjustment for arm preference and opportunities. The data in the third line of Table 2 were adjusted to correct for the number of choices remaining following each correct choice according to the formula

$$\left[\frac{\text{sequential position score for choice}}{\text{sum of sequential position scores for all choices}} \times 100 \right] \times \frac{100}{\text{number of choices remaining}}.$$

The adjusted scores can be 0, indicating that no errors were made to the choice in question, or they can be some positive number, indicating that some errors were made to that choice; the greater the number, the greater the relative number of errors made to that choice. These adjusted scores, which are presented in the fourth line of Table 2, correct for both arm preference and the number of opportunities to repeat each choice and indicate the likelihood of repeating a choice any time an error is made.

Generalization among arms. Table 2 presented information about the sequential characteristics of errors but did not provide information about their spatial characteristics. When errors occurred, they appeared to be randomly distributed among the 8 arms and not preferentially directed near correct arms. An analysis of the spatial distribution of each error was conducted by numbering the position of the arm on which the error occurred as 0, the position of the arms at 45° (immediately adjacent), as 1, the position of the arms at 90° as 2, the position of the arms at 135° as 3, and the position of the remaining arm at 180° as 4. An error location score, indicating the average distance between the repeated arm and each of the remaining correct arms was calculated from the formula

$$\frac{\text{sum of positions of remaining correct arms}}{\text{number of remaining correct arms}}.$$

Error location scores can range from 1.0, indicating that an error was always made to

an arm adjacent to a correct arm, to 4.0, indicating that an error was always made to the arm farthest away from a correct arm. Since there were two arms each at 45°, 90°, and 135°, which were assigned values of 1, 2, and 3, respectively, and one arm at 180°, which was assigned a value of 4, chance performance would be $(2 + 4 + 6 + 4)/7 = 2.29$. The error location scores for Days 1-10 and Days 11-20 ($n = 7$ and $N = 11$) ranged from 2.3 (± 0.03) to 2.5 (± 0.1). These scores closely approximate the values expected by chance, indicating that when errors were made, there was no tendency for them to be to arms close to correct ones.

Effect of errors on choice accuracy. Table 1 demonstrated that there was a general performance decrement as the number of choices increased within a day but did not consider whether the previous choices were correct or incorrect. A correct choice was, by definition, a new choice for the animal, one which had not been made previously on that day. An incorrect choice was a repetition of an earlier choice, and thus the animal made two responses to the same arm. If the rat treats the repetition of an arm as if it were any other choice, then there ought to be an additional performance decrement as a result of the additional choice, reflecting the results reported in Table 1. If the rat treats the repetition of the arm as if it were another instance of the original choice, then there ought to be no additional performance decrement as a result of the error. The question is whether the rat processes the error as a new piece of information, which would result in a performance decrement on subsequent choices or as confirmation of an already acquired piece of information, which might not result in a performance decrement. To answer this question, an analysis was conducted to determine the probability of a correct response on Choice 7 and Choice 8 as a function of the number of prior errors. For each rat, test days were grouped according to the number of errors made prior to Choice 7 for the analysis of choice accuracy on Choice 7 and the number of errors prior to Choice 8 for the analysis of choice accuracy on Choice 8. Only Choices 7 and 8 were analyzed because the frequency of errors

prior to other choices was too low to provide a sufficient number of observations for analysis. The probability of a correct response was then determined by the formula

$$\frac{\text{number of correct responses}}{\text{number of observations}}$$

Scores can range from 0, indicating that none of the observed responses were correct, to 100, indicating that all of the observed responses were correct. As in Table 1, these scores are biased because the chance probability of a correct response varies according to the choice number and the number of previous errors. Consequently, the same correction as applied in Line 2 of Table 1, was carried out here according to the formula

$$\frac{\frac{\text{number of correct responses}}{\text{number of observations}} - p(\text{cor})_{\text{exp}}}{1 - p(\text{cor})_{\text{exp}}}$$

These scores have the same characteristics as those presented in the second line of Table 1.

The results of both analyses indicated that there was no systematic change in the probability of a correct response as a function of the number of preceding errors. The observed trend was in all cases for the probability of a correct response to be lower after an error than after all correct choices, but this trend was not reliable.

Discussion

The results of Experiment 4 provide evidence about the characteristics of the memory processes the rats use while solving the problem. First, the data indicate that there was a progressive decrease in choice accuracy as the number of choices increased (Table 1, Line 1), even when a correction was made for the expected chance level (Table 1, Line 2). Second, when errors occurred, there was no tendency for them not to be made to the arms chosen on the first few choices of the day (Table 2, Line 1), even when corrections were made for the number of opportunities to repeat choices (Table 2, Line 2), arm preference (Table 2, Line 3), and both the number of opportunities to repeat choices and arm preferences (Table 2, Line 4).

Third, when errors occurred, there was a reliable tendency for them not to be made to the arms that had just been chosen (Table 2, Line 1), even when the corrections indicated above had been taken into consideration (Table 2, Lines 2-4), although this tendency was relatively weak. Fourth, there was no indication that the performance decrement as a function of increasing numbers of choices was lessened by an error; indeed, the trend that appeared (which was statistically insignificant) was for an increase in the performance decrement (i.e., a lowered probability of a correct response) after an error. Fifth, there was no indication that the animals confused the different spatial locations and directed their errors toward arms that were adjacent to correct arms.

Two types of corrections were made on the observed data. The first considered the changing chance probability of a correct response and was straightforward. The second considered the possibility of a preference for particular arms, and the assumption was made in Table 2, Lines 3 and 4, that an arm preference did exist. The low correlation between arm preference for Days 1-10 and Days 11-20 raises a question as to the validity of this assumption. One possibility is that an arm preference did exist but was unstable and changed markedly from one block of days to the next. An alternative is that an arm preference did not exist but resulted from the limited sample size which was available for analysis. In any case, the correction for arm preference changes the data in only a quantitative and not a qualitative manner so that the conclusions from this experiment remain unaltered. The absolute magnitude of the tendency not to make errors to recently made choices must remain in question however.

The finding that rats did not tend to make errors to arms close to correct arms suggests that there was no generalization gradient among the arms and that each arm was treated independently of the others. These results appeared to be contradictory to those of Tolman et al. (1946), who reported that in a "sunburst" maze, rats tended to choose arms close to the one that led to the goal. In

that experiment, however, passage down the correct arm was blocked. In the present experiment, of course, access to all arms was available at all times. Thus, the paradigms are substantially different, and the results cannot be directly compared.

EXPERIMENT 5

Experiment 4 demonstrated that there was a performance decrement as the number of choices increased (see Table 1) but left unresolved the issue of whether this decrement was due to the number of choices or just the passage of time. Experiment 5 was designed to answer this question by forcing the animals to spend time in the center compartment between the fourth and fifth choices. If the performance decrement in Experiment 4 was due simply to the passage of time, then performance on Choices 5-8 should be impaired after confinement. If the performance decrement was due to the choices themselves, then no impairment should be observed.

Method

Subjects and apparatus. The subjects were six of the animals from Experiment 4 maintained under the same conditions. The apparatus was the same as in Experiment 4.

Procedure

The procedure was the same as in Experiment 4 except that animals were confined to the center platform after the fourth choice. One test was given each day for 5 days. When the animal returned to the center platform following the fourth choice, the guillotine doors were lowered for at least 2 min., more than the amount of time that the animals usually required to complete the last four choices. Following this confinement, the guillotine doors were raised and the animals allowed to choose freely until the remaining four reinforcements were obtained.

Results

All animals readily adapted to the confinement procedure. During the confinement period, the animals were active and moved about the center compartment in the typically exploratory fashion of rats. When the guillotine doors were raised, the rats usually proceeded to the nearest arm and either chose that arm or moved to some other arm. There was no obvious change in the accuracy of

performance as a result of the confinement procedure, and the mean of the number of different arms chosen within the first 8 choices was 7.5 (range: 7.2–7.9) as compared with 7.6 (range: 7.3–8.0) during Days 11–20 of the previous experiment.

Discussion

Interference theories of forgetting propose that information is lost because new information interferes with the storage or retrieval of previous information and not simply because of the passage of time (Hulse et al., 1975). The current evidence supports the notion that the decrement in performance observed here occurs because of the increased number of choices and not because of the time interval between previous choices and the current choice. The period over which rats can successfully remember their previous choices was not addressed by the present experiment, but is obviously much longer than the few minutes tested here and may approach the limits found in spontaneous alternation experiments (Douglas, 1966; Walker, 1956).

EXPERIMENT 6

The results of Experiment 4 demonstrated that there was a performance decrement as the number of choices increased on each day but that this decrement was alleviated between days. These results indicate that the animals were able to successfully separate each day's performance from the previous day's performance so that choice accuracy recovered to its original level at the start of each day. The present experiment was designed to determine how rapidly recovery of performance could take place. To this end, the rats were allowed to choose all eight arms and were then confined to the center platform while all eight arms were rebaited. They were then allowed to choose all eight arms again. The results suggest that the animal could successfully "parse" their learning under these conditions so that there was almost complete recovery of choice accuracy between the last choice of the preceding series and the first choice of the following series.

Method

Subjects and apparatus. The subjects were four rats of Experiment 4 maintained under the same conditions as described previously. The apparatus was the same as in Experiment 4.

Procedure

The procedure was the same as in Experiment 4 with the following exception: After completing choices of each of the eight arms, the animals were confined to the center platform by the guillotine doors. All arms were rebaited, a process which required about 1 min., the guillotine doors raised, and the animals allowed to choose freely until all eight arms were chosen again. This procedure was then repeated until the animal was given a total of four tests on each of the first 5 days and a total of eight tests on each of the next 5 days.

Results and Discussion

On the first day of repeated tests, the animals chose normally on the first test but tended to choose slowly on the subsequent tests. On the following days, the animals readily adapted to the procedure and chose at their normal rate on all tests. There was a slight but consistent decrease in choice accuracy from the level demonstrated on the first test of the day, but even on the first day that the animals were given four tests, choice accuracy on the last choice was still greater than that observed on the first 5 days of Experiment 4. The same results were found when the animals were given eight tests on each day, demonstrating that the rats were able to successfully separate each test in memory so that the performance decrement expected as a function of the increasing number of choices did not appear. Table 3 presents these data.

GENERAL DISCUSSION

The general testing paradigm described here may be summarized as sampling with replacement from a known set of items until the entire set is sampled. The cognitive requirements of the present paradigm are straightforward; the animal must be able to locate each of the eight different arms and determine whether each arm has been chosen. Numerous strategies are possible to solve the task. The easiest approach would be to

TABLE 3
PERFORMANCE ON CONSECUTIVE TESTS

Test	Four tests				Eight tests			
	First day		All days		First day		All days	
	<i>M</i>	<i>SEM</i>	<i>M</i>	<i>SEM</i>	<i>M</i>	<i>SEM</i>	<i>M</i>	<i>SEM</i>
1	7.3	.3	7.8	.1	7.8	.1	7.9	.04
2	7.3	.2	7.4	.3	8.0	.0	7.6	.1
3	6.8	.4	6.9	.1	7.5	.2	7.5	.03
4	6.8	.3	6.9	.1	7.3	.4	7.5	.0
5					6.8	.4	7.3	.2
6					7.0	.3	7.5	.2
7					7.5	.2	7.4	.2
8					7.5	.2	7.3	.1

Note. Table entries are the mean and standard error of the mean of the number of correct choices with the first eight choices of each test.

choose adjacent arms; the choice strategy could be stored in long-term memory, and the arms could be responded to as they appeared and would not have to be individually identified or remembered. A slightly more complicated approach would be to mark each arm as it is chosen; the marking system could be stored in long-term memory and choices guided by the number of marks present at each arm. Many other types of strategies are available as well. But the data indicate that the rats did not choose adjacent arms, did not rely on any type of intramaze marking, did not chain responses in particular sequences, and did not utilize any other obvious strategy to simplify the task. Since the testing procedure involves sampling with replacement, the rats must have been able simultaneously to identify each of the eight arms and remember over a period of several minutes which arms had been chosen and which had not.

Further analysis of the cues used by rats was not undertaken in the present experiment because it was not relevant to determining the general type of strategy used in solving the problem. Previous research has indicated that rats in maze tasks utilize almost any cue available to them (see discussion by Kimble, 1961; Woodworth, 1938). Such is probably also the case here, but accurate evaluation of the relative importance of different extramaze, directional, or compass cues requires a major experimental effort and is beyond the scope of the present paper.

Other experiments, although slightly dif-

ferent from the present one, have used a procedure closely resembling sampling with replacement from a known set of items until the entire set is sampled. The most similar animal experiment is that of Menzel (1973). Monkeys first observed 18 food items being distributed around a large enclosed compound and were then released to search for the food. They obtained an average of 12.5 food items and did so in a manner which minimized the distance which had to be traveled. The most similar human experiment is the "missing scan" technique of Buschke (1963). Subjects were successively presented with all but one item from a known list and then asked to identify the item that was missing. They had no difficulty in performing accurately with lists of up to 12 items. Thus, rats, monkeys, and humans all demonstrate a substantial memory capacity when tested in a sampling with replacement paradigm. An interspecies comparison of the limits of the memory capacity is not yet possible because of differences in procedure among the experiments and because none of the experiments was difficult enough to substantially challenge the subjects' capabilities. Other experiments (Dennis, 1939; Hunter, as analyzed by Miller & Frick, 1949; Tinkelpaugh, 1932; Yerkes & Yerkes, 1928) have also addressed the issue of memory for multiple events, but the paradigms were different enough from the present one to prevent direct comparison of results.

The rats rapidly attained almost perfect performance. Even performance on the first few days was far above the level expected on

the basis of mathematical chance. But mathematical chance assumes that all choices are independent of each other—an assumption which is undoubtedly invalid. Other experiments (Olton, Note 1) have demonstrated a substantial spontaneous alternation tendency (Douglas, 1966) in the present situation; when all choices of all arms were reinforced, rats still chose an average of approximately six different arms within eight choices, a performance level which is very similar to that found here during the first 5 days of the experiment. Thus, even under conditions when reinforcement contingencies do not encourage choosing all arms, substantial alternation still occurs. There are several conclusions to be drawn from these data: First, the rats bring to the experimental situation a strong tendency to perform the task correctly. This is presumably a reflection of spontaneous alternation. Second, the reinforcement contingencies produce an improvement in the number of correct choices, and animals learn to perform better after training. Third, the presence of a spontaneous alternation tendency influences only the baseline level of performance exhibited by the rats and not the strategy used to solve the task; during spontaneous alternation, rats do not choose adjacent arms, chain responses, or utilize intramaze cues (Olton, Note 1).

The sequential position of errors analysis indicated that the tendency to make an error by repeating a response to a particular arm increased with the number of responses made after that arm had been chosen, even when corrections were made for arm preference and for the number of opportunities to repeat each choice. Some of this tendency may have been due to the physical constraints of the apparatus because animals returning to the center platform after choosing an arm were directed away from the arm just chosen. Any such effect would be very minor however. First, the rats often did not go directly to another arm after returning to the center platform; rather, they explored the entrance to several arms, often completing one or more revolutions around the center platform before making a choice. Second, even if the rats did tend to respond to arms physically

opposite the one just chosen, following the subsequent choice they would be facing back toward the arm they had just previously chosen. Third, an analysis of response sequences indicated that although there was a slight preference for arms at 90° from the chosen arm, such preferences were weak and not a prerequisite for accurate performance. Thus the tendency for errors to be made by repeating choices early in the day's test was robust and reliable and does not appear to be due to the physical characteristics of the apparatus.

Taken together, these data suggest several conclusions about behavior in the present paradigm. First, the animals had no difficulty identifying the eight spatial locations and remembering whether they had chosen each one; although performance was not perfect, even on the eighth choice it was considerably above chance. Second, there was a small but consistent decrement in performance as the number of choices increased; this decrement was due to interference from previous choices and not just the passage of time. Third, analysis of the effects of prior choices on the probability of an error indicated that the tendency to make an error by repeating a choice was a function of the number of choices since the original correct choice. Fourth, analysis of the effects of prior choices on the location of errors indicated that the eight arms were treated as a separate place and there was no apparent tendency for spatial generalization among arms. Fifth, information was stored choice by choice and repetitions of a choice (i.e., errors) were stored in the same manner as a new choice. Sixth, rats were able to store information from sequential tests separately so that there was no performance decrement of one test on following tests.

Although the present paradigm is operationally different from the usual human paradigms in which primacy, recency, proactive interference, and retroactive interference have been studied, it is nonetheless conceptually very similar. Thus the primacy-recency analysis of human data is similar to the sequential location of errors analysis presented in Table 2. The equivalent of the primacy effect in human serial learning

would appear in the present analysis as a lowered probability of an error to choices made early in the test, while the equivalent of a recency effect would appear as a lowered probability of an error to choices made near the end of the test. No indication of a primacy effect was found in any of the analyses. Data from human subjects suggests that the primacy effect may result from more rehearsal of items at the beginning of the list than in the middle; accuracy of recall is a function of the amount of rehearsal, which is greatest for items in the beginning of the list (Rundus, 1971), and when subjects are instructed to rehearse only the most recent item, the primacy effect does not appear (Waugh & Norman, 1965). If the absence of a primacy effect for rodents is a consistent finding, it may suggest that rats do not rehearse their choices or that rehearsal is not preferentially directed toward early choices, in the tasks involved, but equally distributed among all choices. Alternatively, the primacy effect in the human data may be due to the requirement of serial ordering of items and the formation of chains of responses (Mandler & Anderson, 1971). In the present paradigm, of course, rats were not required to chain responses and no particular sequences of choices were preferred over others. Since humans almost inevitably order responses, even when asked to randomize them, a test of this hypothesis can probably be conducted only by testing rats (or some other animal that does not automatically sequence responses), forcing them to choose in a particular serial order and then determining if a primacy effect appears.

The equivalent of proactive and/or retroactive interference in human serial learning would appear in the present analysis as a decrement in performance as the number of choices increased. There was a reliable decrement in performance as the number of choices increased during the day's test, and this decrement appeared even when a correction had been made for the varying chance probability of a correct response. The present experiment is unable to identify definitively whether the performance decrement observed here is due to proactive or retroactive interference because the way in which

the animals remember which arms have been chosen and/or which arms have not been chosen has not yet been determined. Data from discrimination learning experiments strongly suggest that rats perform discriminations largely on the basis of nonreinforcement (see discussion in Olton, 1972; Olton, Walker, Gage, & Johnson, Note 2). If such is the case in the present experiment, then rats probably remember not to return to arms which have already been chosen because these would result in nonreinforcement if chosen again. The performance decrement would then be due to retroactive interference. Other evidence also supports the suggestion of retroactive rather than proactive interference. The finding of a recency effect in the absence of a primacy effect suggests that the animals are remembering where they have been and which arms ought not to be chosen. The fact that animals could successfully overcome the expected performance decrement when given four or eight tests on each day is also compatible with the suggestion of retroactive interference.

In summary, the results of the present experiments are important because they demonstrate that rats have a remarkable memory capacity for information about spatial location and because they provide information about some of the characteristics of the storage process used in this memory. The data also suggest that just as the introduction of the "interoceptive" learning paradigm (Garcia, Hankins, & Rusiniak, 1974; Rozin & Kalat, 1971) substantially changed our estimate of rats' learning abilities, so the introduction of a spatial location paradigm may change our estimate of rats' cognitive capabilities.

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APPENDIX

TABLE A1

DAILY SEQUENCE OF FIRST EIGHT CHOICES OF RAT 3 IN EXPERIMENT 1

Day	Order	Day	Order	Day	Order
10	85726354	16	76812453	22	76538124
11	85732615	17	83521747	23	86574123
12	58317246	18	86457138	24	64572831
13	68745723	19	45682375	25	46351872
14	73654128	20	63817425	26	46273518
15	75436812	21	86457371	27	84252371

Note. The numbers in the order columns refer to Arms 1-8 of the radial maze.

(Table A2 is on the next page.)

TABLE A2
RAW DATA FROM DAYS 1-10 IN EXPERIMENT 4

Rat	Day	Order	Rat	Day	Order	Rat	Day	Order
1	1	76512678365824	5	1	181275317416	9	1	248842137156
	2	71237463845		2	47186325		2	181267478253
	3	75567823451		3	25386417		3	212564718537
	4	67543218		4	16428537		4	73512864
	5	37812456		5	15387264		5	713682514
	6	5678134562		6	1474825136		6	71326485
	7	32187654		7	17534826		7	875136224
	8	35781246		8	14283657		8	15372684
	9	16373135675428		9	73528461		9	28637514
	10	2187573456		10	52831746		10	615247328
2	1	671348625	6	1	75861517356842	10	1	1854231514367
	2	783624715		2	18763518462		2	5724543186
	3	8573246851		3	14678253		3	853658146318572
	4	74568123		4	13457826		4	452641873
	5	687413245		5	67824513		5	35784612
	6	4632571478		6	78135624		6	74318562
	7	67824513		7	78246135		7	17542186
	8	68345712		8	68245713		8	57132468
	9	67814523		9	71352864		9	48154321876
	10	72345186		10	74623815		10	58261437
3	1	7878351246	7	1	76145317528	11	1	2182617354
	2	418514687356142		2	7163816542		2	74318562
	3	712856134		3	4637154218		3	58416273
	4	14536827		4	132518476		4	461742583
	5	42537146217348		5	1358254712356		5	18537462
	6	12437518425417836		6	71435628		6	7685137264
	7	24682467351		7	41725836		7	76845132
	8	24782513476		8	43682571		8	768471523
	9	634671378245		9	58167243		9	8675164432
	10	31753885264		10	57123468		10	68471565723
4	1	18431748751826	8	1	85326487	12	1	1825175346
	2	32164871864325		2	2874173685		2	187428651413
	3	45231786		3	86174318652		3	18178235864
	4	86431752		4	851748623		4	1541265783
	5	48632175		5	86231547		5	613852147
	6	32876541		6	81642753		6	816432567
	7	16572843		7	68725314		7	7625813725814
	8	53286417		8	86571423		8	5686413712
	9	48762513		9	86372541		9	53461248217
	10	65482173		10	86125473		10	31785426

Note. The numbers in the order columns refer to Arms 1-8 of the radial maze.

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